

2024



AgriProve Uses AWS and EOSDA to Help Farmers Improve Soil and Address Climate Change

Using a satellite-based precision agriculture solution built on AWS by EOS Data Analytics Inc., carbon soiltech developer AgriProve increases the efficacy, impact, and reporting of projects.

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baseline reports using accurate histori



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Reduced time



Australian company [AgriProve](#) uses Amazon Web Services (AWS) and [EOS Data Analytics Inc.](#) (EOSDA) technologies to help transform Australia's agricultural practices by assisting farmers in producing healthier soil and operating more productive farms. This collaboration helps accelerate progress toward a net negative carbon economy, combating climate change and revitalizing critical ecosystems.

Practices like soil carbon farming land management and regenerative agricultural increase carbon storage in soil and soil health, improve farm yield, remove carbon from the air, and reduce greenhouse gas emissions. Healthier soil unlocks multiple benefits, including land that is more productive and resilient to Australia's harsh weather conditions, which in turn helps farmers to access additional revenue. With many farmers keen to move beyond net zero as part of a commitment to tackling climate change, the appetite to harness more data continues to grow.

AgriProve is a pioneer in harnessing technology, innovation, and data to help land managers improve soil health. AgriProve manages more than 600 projects across 165,000 hectares using a data-driven approach that integrates technological innovation with commercial and ecological integrity to support farmers.

AgriProve's work is rewarded with carbon credits from the Australian government. For example, in 2019, AgriProve's Gippsland soil carbon project was the first project to be issued Australian carbon credits under the Australian government's Emissions Reduction Fund.



Saving Time and Increasing Data Analysis Accuracy Using AWS and EOSDA

In 2021, AgriProve collaborated with EOSDA, a global provider of satellite imagery analytics, to provide data and imagery support and assist in generating soil organic carbon models. Founded by Dr. Max Polyakov, EOSDA uses artificial intelligence and proprietary algorithms to derive geospatial data and analyze satellite imagery for business and science uses and to make a positive change.

"We want to be able to show that the land management strategy we're putting in place at the beginning of a project is going to generate additional containment of carbon. The historical information and analytics that we get from EOSDA Crop Monitoring enhanced our ability to do that," said Matthew Warnken, managing director at AgriProve.

[EOSDA Crop Monitoring](#), an online satellite field monitoring service, is fully built on AWS, including infrastructure for processing, compute, storage, and data transfer. Customers like AgriProve use the EOSDA Crop Monitoring solution to analyze historical data, create baseline reports, and track the farmland's progress throughout a project.

EOSDA uses [AWS Lambda](#) serverless compute to help process information collected from satellite data and [Amazon Simple Storage Service](#) (Amazon S3) to store large volumes of data. In 2022, EOSDA made several of its solutions, including EOSDA Crop Monitoring, available through the [AWS Marketplace](#), where organizations can find, test, buy, and deploy software that runs on AWS.

AgriProve reports how the land has been worked in the past and how it responds to regenerative land management practices so that a project can be registered and continue to be compliant with the Australian regulatory framework. This enables AgriProve to share quantifiable soil carbon trends on a regular basis. A five-year retrospective baseline carbon benchmark is required to register projects and repeat [benchmarking](#) is needed throughout the entire 25-year-plus project life cycle.

"The analytics from EOSDA help AgriProve put forward projects with even more rigor and science behind them while making it even easier for the farmers we work with to see how and where what they do with their land is boosting soil health," said Warnken.

Using EOSDA Crop Monitoring data analysis powered by AWS, farmers and landholders can better understand, track, and benchmark the impact that their different land management techniques—including grazing intensity and frequency. Of note, this is required to participate in the Australian Carbon Program to avoid the release of greenhouse gas emissions or remove carbon from the atmosphere. It also helps increase productivity and yield as a result of healthier soil.

Making Carbon Credit Verification more Affordable for Farmers

AgriProve continues to invest in EOSDA Crop Monitoring and use the data to improve its processes, project efficacy, and customer experiences. Since 2022, EOSDA and AgriProve also have collaborated on the Australian government's National Soil Carbon Innovation Challenge to use synthetic aperture radar and optics to improve soil carbon storage and measurement.

By developing this technology and leveraging AWS, EOSDA and AgriProve aim to reduce the number of physical samples needed for Australia's Emissions Reduction Fund to verify carbon credits. This would significantly reduce the cost of establishing and running a carbon project and make it more affordable for farmers. The Emissions Reduction Fund provides businesses the opportunity to earn Australian carbon credits for every ton of carbon dioxide and avoids emissions using new practices and technologies.

"Innovation has always been key for us," said Ferhod Sarmed, chief technology officer at AgriProve. "On AWS, EOSDA has been with us on our journey and is there to support us with innovation and accurate information."

About EOSDA

EOS Data Analytics Inc. (EOSDA) offers satellite imagery analytics worldwide. Its EOSDA Crop Monitoring solution is a one-stop shop that includes multiple types of data—including crop health, weather conditions, crop rotation, field activities, elevation, soil moisture, and more—all in one place.

About AgriProve

AgriProve is a leading carbon soiltech developer with over 70 percent of all soil carbon projects successfully registered in the Australian Carbon Credit Unit Scheme. It has over 600 projects totaling more than 165,000 hectares as of November 2023.

AWS Services Used

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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AWS Lambda



AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

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AWS Marketplace

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Global seed producer KWS uses Amazon Bedrock and generative artificial intelligence (AI) to boost research speed and efficiency. Supported by Adastra, an Amazon Web Services (AWS) Partner, KWS can explore new use cases powered by different large language models, knowing it's operating securely and cost-effectively.

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Abit Drives New Age of Sustainable Agriculture...

Abit's artificial intelligence (AI) prediction systems help farmers and businesses combat climate change by better understanding and supporting biodiversity in agriculture. All-in on Amazon Web Services (AWS), the startup uses generative AI and Amazon



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Indigo Ag Uses Amazon Aurora to Scale...

Agriculture technology provider Indigo Ag began to migrate its services to Amazon Aurora for a simple, scalable, fully managed solution that supports growth while decreasing work for its developers.

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CropX Runs Soil Sensor Application on AWS to...

CropX uses an AWS-based big data solution to enable agricultural sustainability, support a fourfold increase in soil sensor data, quadruple annual sales, and give farmers new insights into their crops.

2020

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